Supplementary Material: What Does Deduplication Delete? A Fact-Level Framework for Auditing Information Loss in Web Corpora

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References to sections, figures, and tables without an "S" prefix point to the main paper.

S1 Sampling and Analysis Populations

AP expansion. The expansion samples NewsEdits AP entries rather than individual versions. Of 12,985 entries with multiple versions, exclusion of previously used or exposed entries leaves 12,706 candidates. Versions are ordered by creation time, version, and identifier. For entries with more than six versions, six equally spaced ranks, including the endpoints, are retained. HTML removal and entity decoding precede exclusion of bodies shorter than 200 characters. Families with fewer than two remaining versions are excluded, leaving 12,700 candidates. Proportional allocation by retained version count uses the largest-remainder rule and seed 20261009:

Versions	Candidate families	Sampled families
2	3,314	235
3	1,332	94
4	1,147	81
5	1,520	108
6	5,387	382
Total	12,700	900

Of the 900 sampled families, 898 yield usable extraction and complete support matrices. Reusing 119 frozen families gives the final 1,017-family panel. Failures are excluded without replacement. The earlier sample used dependency-based families spanning 133 entries, whereas the expansion uses entry-based families. Entry identifiers do not overlap across the samples; story-level independence across entries was not established. Differences between old and new estimates therefore mix sampling and grouping changes.

FineWeb expansion. The input comprises a fixed-order selection of 1,500,000 new unique page identifiers from the same crawl as the earlier 100,000-page pool. Existing downloaded rows are considered first, followed by Parquet files and their row order. This is not a random sample of the crawl. Discovery uses full-text word 5-grams, 64 MinHash permutations, and 16 bands of four hashes. Candidate pairs with exact Jaccard similarity at least 0.3 form single-linkage families; LSH buckets larger than 50 pages do not generate candidates. This discovery step is distinct from within-family policy evaluation, which computes exact shingle Jaccard similarities.

The new pool yields 4,607 non-singleton families. Excluding 58 families connected to previously used families and 53 failing the short-text condition leaves 4,496 candidates. With seed 20261009, proportional largest-remainder allocation by full family size selects 900 families. Each contributes at most four pages with the smallest input indices, each truncated to 6,000 characters. Six selected families exhaust extraction attempts with empty inventories; 894 usable families plus 115 frozen families form the final 1,009-family panel. Discovery counts describe this retrieval procedure's detections, not duplicate prevalence with complete recall. Because the published FineWeb inputs have already undergone deduplication, this panel concerns residual duplicates.

CC-News. The frozen public CC-News input contains 708,241 articles. Full-text discovery uses the same shingle, MinHash, LSH, Jaccard, and oversized-bucket rules described above. Of 71,854 non-singleton families, 46,488 span at least two known normalized hostnames; 45,893 satisfy the sampling conditions. Seed 20261010 and proportional largest-remainder allocation by full family size select 500 families. Within each, selection prioritizes the earliest article, the latest, and another hostname, then fills remaining slots in order, up to four articles. Bodies shorter than 200 characters are ineligible; selected bodies are truncated to 6,000 characters. The panel contains 1,436 articles, of which 98 are truncated. Rare large families may receive no sampling quota. Hostnames retain subdomains and do not identify independent publishers.

If any selectable member of a full family lacks a date, the procedure substitutes row order for the entire family. This affects 73/500 sampled families; the order is not a verified publication chronology. Date-complete families with distinguishable timestamps form a 307-family sensitivity subset. One short-news family yielded only three facts after repeated extraction; its documented exception retains the final output without changing prompts or replacing the family. The report also provides a sensitivity analysis excluding it. All 500 families have complete support matrices. A frozen split assigns 350 families to development and 150 to matched-budget testing.

Denominators. The final AP and FineWeb panels contain 42,433 and 24,946 non-outdated facts, respectively. Loss rates pool fact counts rather than average family percentages. Removal rate always divides removed words by exact-deduplicated words: 1,672,964 for AP, 665,272 for FineWeb, and 445,778 for CC-News. Exact text identity lowercases the concatenated title and body and collapses whitespace; word counts use whitespace-delimited body tokens. The inventories and judgments cover selected texts only. Family membership does not establish that a fact is absent elsewhere in the retained corpus.

S2 Calibration and Sensitivity Analysis

Human evidence and target population. The final calibration combines 250 first-round single-annotator judgments with 240 second-round final human labels, yielding 490 cells. Two annotators independently agreed on 228/240 second-round cells (95.00%; Cohen's $\kappa = 0.9207$). They jointly adjudicated the remaining 12 cells without seeing the model labels. Agreement between the model and the final human labels is 215/240 (89.58%). The second-round sampling frame comprises eligible frozen families, bodies of at most 900 whitespace-delimited tokens, and deduplicated fact–text cells,

excluding prior annotations and identical content. Its 192 original-text cells and 48 Sub-doc merge cells measure cell-level agreement, not error-free documents.

Sampling strata cross panel, text source, coverage, and model label. Design-weighted agreement uses $w_h = N_h/n_h$, where N_h and n_h denote eligible and sampled cells in stratum h. Its final value is 96.72%, with total eligible-cell weight 183,844. This descriptive weighting differs from calibration: transition posteriors use unweighted observed counts conditional on panel, original versus merged text, and model label. Population weights are not treated as independent observations. The first round includes 30 targeted disagreement cells, motivating an exclusion sensitivity analysis.

Sequential conditional Jeffreys model. For an observed model label g, define the first alternative a as N when g is S, and S when g is N or C. Let b be the other alternative. Within a conditional row, n is the observed cell count, and k_a, k_b, k_g count the human labels. Draw

$$q_1 \sim \mathrm{Beta}(k_a + \tfrac{1}{2}, n - k_a + \tfrac{1}{2}),$$
$$q_2 \sim \mathrm{Beta}(k_b + \tfrac{1}{2}, k_g + \tfrac{1}{2}).$$

The probabilities assigned to a, b, g are respectively q_1, $(1 - q_1)q_2$, and $(1 - q_1)(1 - q_2)$. Each conditional Bernoulli parameter has a Jeffreys prior; this is not a symmetric three-class Dirichlet model. An unobserved transition retains posterior uncertainty, especially in the small merged-text strata.

Shared-text posterior prediction. The sampling unit for a predicted label is the tuple (family, fact index, normalized title and body). All occurrences of that unit across original copies and policy outputs share one sampled label. Original-text labels take precedence when identical text appears in both original and transformed outputs; otherwise frozen output order resolves duplicate labels. This is a computational consistency rule, not a new human adjudication. Distinct original and merged texts use their respective error posteriors. The rule prevents identical retained text from losing support merely because duplicate occurrences receive independent flips.

For each draw, the perturbed original support matrix determines whether each fact is valid and whether it is non-outdated or outdated. All policies use these recomputed categories and the same denominators within the draw. A fact must have original support to be valid; contradiction in the latest source determines the outdated category. Calibration therefore recomputes fact denominators within the fixed inventory.

Joint uncertainty. The final analysis uses 2,000 posterior-predictive rounds per scenario with seed 20261014. Each round also resamples families with replacement independently within AP and FineWeb. All policies share the same family weights, and repeated copies of a family reuse its sampled labels. Joint sensitivity intervals take the 2.5th and 97.5th percentiles of the resulting distribution; fixed-family, label-only intervals are retained separately. Matched-budget comparisons keep the frozen allocation, rather than reallocating words after each bootstrap draw. The fraction of rounds with a positive difference is neither a p-value nor an independent replication success rate.

Scenarios and interpretation. The final main scenario uses all 490 cells. Excluding the 30 targeted first-round cells leaves 460. A further scenario changes the error posterior borrowed for previously unseen sentence-level outputs from merged-text errors to original-text errors; identical texts still share labels. The earlier calibration analysis instead used 220 non-targeted original-text cells, with separate-panel and pooled-S scenarios. Its transfer of original-text errors to new merged outputs was not independent validation of those outputs. Historical pre-adjudication scenarios also assigned all unresolved cells to S, N, or C in turn; these do not replace the final adjudicated labels and are not bounds over every mixed assignment.

The three prespecified comparisons retain their directions in 2,000/2,000 rounds in both the final and targeted-cell-exclusion scenarios: AP MinHash-Earliest@0.5 versus MinHash-Latest@0.5; FineWeb MinHash-Earliest@0.3 versus AP MinHash-Latest@0.3; and AP Keep-One-Earliest versus Keep-One-Longest. The final Sentence-Exact minus Sub-doc merge loss differences are 2.415 percentage points for AP (95% CI [−0.221, 12.988]) and 0.586 for FineWeb ([−3.178, 5.334]), at respective removal budgets of 30% and 20% over exact-deduplicated words. These comparisons use 60 AP and 58 FineWeb test families. Both intervals cross zero, leaving the ranking of these sub-document comparators unresolved.

The transition posterior treats calibration cells as conditionally independent, while the evaluation-family bootstrap resamples audited families. The resulting intervals quantify sensitivity under assumed error transfer across text lengths and output types. They leave extraction omissions, annotator mistakes, dependence within the annotation sample, and cross-family story or site dependence outside their scope.

S3 CC-News Results

CC-News combines high fact overlap with an uncertain ordering of representative and threshold effects. We report pointwise 95% family-bootstrap intervals unless a multiplicity correction is specified. Its mean within-family, pairwise Jaccard overlap of supported facts is 91.01% (500 families; 1,542 valid document pairs), with a 95% CI of [89.67, 92.23]%. Below, we report the complete policy table, ordering sensitivities, and matched-budget comparisons.

Panel and denominators. The panel contains 500 families and 1,436 selected documents. Exact deduplication leaves 1,134 texts and 445,778 body words. Of 14,932 extracted facts, 14,910 have source support: 14,867 enter the non-outdated fact inventory and 43 enter outdated retention. Non-outdated facts have support in at least one source and no contradiction in the last ordered source; outdated facts have source support and a contradiction in that last source. Table 1 reports removal relative to exact-deduplicated words, non-outdated fact loss, and outdated retention. Keep-all and Exact have zero removal on this accounting baseline. We use 2,000 family-bootstrap replicates with seed 20261010.

Representative and threshold effects. Let $L_{e,t}$ and $L_{l,t}$ denote loss for MinHash-Earliest and MinHash-Latest at threshold t. The representative effect is $R = [(L_{e,0.3} - L_{l,0.3}) + (L_{e,0.5} - L_{l,0.5})]/2$; the threshold effect is $H = [(L_{e,0.3} - L_{e,0.5}) + (L_{l,0.3} - L_{l,0.5})]/2$. CC-News yields $R = 0.50$ pp [95% CI: −0.50, 1.44], $H = 1.71$ pp [1.08, 2.43], and $R - H = -1.20$ pp [−2.61, 0.08]. Bonferroni 97.5% intervals for the two designated comparisons are [−0.68, 1.54] for R and [−2.79, 0.23]

Table 1: CC-News main policies on all 500 families. Each rate is in percent; brackets give 95% CIs. Removal uses 445,778 exact-deduplicated words; fact columns include their respective counts and denominators.

Policy	Removal [95% CI]	Lost / inventory; rate [95% CI]	Outdated retained / total; rate [95% CI]
Keep-all	0.00% [0.00%, 0.00%]	0/14867; 0.00% [0.00%, 0.00%]	43/43; 100.00% [100.00%, 100.00%]
Exact	0.00% [0.00%, 0.00%]	0/14867; 0.00% [0.00%, 0.00%]	43/43; 100.00% [100.00%, 100.00%]
MinHash-Earliest@0.9	25.40% [21.99%, 28.71%]	80/14867; 0.54% [0.39%, 0.71%]	42/43; 97.67% [91.38%, 100.00%]
MinHash-Latest@0.9	25.45% [22.02%, 28.75%]	75/14867; 0.50% [0.35%, 0.67%]	37/43; 86.05% [69.69%, 98.11%]
MinHash-Earliest@0.8	37.67% [34.65%, 40.63%]	249/14867; 1.67% [1.35%, 2.04%]	42/43; 97.67% [91.38%, 100.00%]
MinHash-Latest@0.8	37.52% [34.49%, 40.50%]	164/14867; 1.10% [0.87%, 1.35%]	27/43; 62.79% [39.28%, 83.33%]
MinHash-Earliest@0.7	44.72% [41.88%, 47.37%]	396/14867; 2.66% [2.21%, 3.14%]	42/43; 97.67% [91.38%, 100.00%]
MinHash-Latest@0.7	44.68% [41.79%, 47.31%]	324/14867; 2.18% [1.79%, 2.61%]	23/43; 53.49% [28.56%, 75.56%]
MinHash-Earliest@0.5	51.70% [49.34%, 53.84%]	670/14867; 4.51% [3.84%, 5.23%]	41/43; 95.35% [86.67%, 100.00%]
MinHash-Latest@0.5	51.32% [48.96%, 53.52%]	543/14867; 3.65% [3.04%, 4.34%]	13/43; 30.23% [5.88%, 55.10%]
MinHash-Earliest@0.3	55.13% [52.94%, 57.17%]	872/14867; 5.87% [4.99%, 6.78%]	41/43; 95.35% [86.67%, 100.00%]
MinHash-Latest@0.3	55.13% [52.95%, 57.20%]	849/14867; 5.71% [4.64%, 6.96%]	1/43; 2.33% [0.00%, 7.15%]
Keep-One-Earliest	56.45% [54.38%, 58.38%]	955/14867; 6.42% [5.40%, 7.53%]	37/43; 86.05% [65.30%, 100.00%]
Keep-One-Latest	56.39% [54.28%, 58.30%]	972/14867; 6.54% [5.18%, 8.03%]	0/43; 0.00% [0.00%, 0.00%]
Keep-One-Longest	54.11% [52.04%, 55.96%]	486/14867; 3.27% [2.46%, 4.24%]	15/43; 34.88% [15.52%, 60.98%]
Keep-One-Random	56.50% [54.45%, 58.42%]	989/14867; 6.65% [5.42%, 8.05%]	27/43; 62.79% [36.84%, 81.82%]

for $R - H$. Thus, high overlap accompanies small losses without establishing a tuning priority in these signed comparisons.

Date sensitivity. Missing dates require an ordering fallback in 73/500 families, covering 227 documents. These families use row order throughout, rather than mixing rows with timestamps. Among the 307 families with complete, distinct dates, the non-outdated denominator is 9,385 facts: $R = 0.75$ pp [95% CI: $-0.45, 2.07$], $H = 2.26$ pp $[1.38, 3.26]$, and $R - H = -1.51$ pp $[-3.19, 0.13]$. Restricting the analysis to this subset leaves the tuning priority unresolved.

Low-fact-family sensitivity. One short-alert family produced only three facts, below the requested range of 8–45; the exception was recorded before its final judgment and aggregate analysis. Excluding it leaves 499 families and 14,864 non-outdated facts. Main-policy loss estimates change by at most 0.001343 pp and designated effect estimates by at most 0.000415 pp, with $R = 0.5046$ and $H = 1.7088$ pp. The uncorrected $R - H$ interval becomes $[-2.50, -0.03]$ pp, while its 97.5% corrected interval remains $[-2.63, 0.13]$ pp. This family belongs to development, so the matched-budget test results are unchanged; the boundary crossing under resampling leaves a robust threshold-priority claim unsupported.

Matched removal budgets. The split contains 350 development and 150 test families. The test baseline is 140,636 exact-deduplicated words, with 4,483 non-outdated and 10 outdated facts. All six methods meet the rounded integer removal targets of 14,064 and 28,127 words for nominal 10% and 20% budgets. Development selects candidates using word counts; a multiple-choice subset-sum controller assigns those candidates or the identity output to test families without reading fact loss. Latest and Longest in Table 2 select representatives within document single-linkage components. Sentence-Exact traverses sources newest first and keeps the first normalized sentence occurrence. Sub-doc merge retains the latest representative and appends sentences with Jaccard below 0.5 against its original sentences, without deduplicating appended sentences against each other. The C4-style and Dolma-style transformations follow Appendix S4. Sentence-Exact can remove at most 41,900/140,636 words (29.7932%) with the frozen candidates, so a 30% joint budget is unavailable.

Sentence-Exact has favorable loss point estimates against Latest at both budgets, but all four comparisons against Latest and Longest have intervals crossing zero (Table 3). Sub-doc merge loses fewer facts than Latest in the two pointwise comparisons, while its comparisons with Longest and Sentence-Exact cross zero. C4-style and Dolma-style have similar point estimates to Sentence-Exact (Table 4). These model-label comparisons lack new human calibration and multiplicity correction.

S4 C4-style and Dolma-style Comparisons

C4-style captures much of Sentence-Exact's loss reduction under the primary small-panel allocation, while Dolma-style is sensitive to repeated judging and allocation. These family-local adaptations isolate mechanisms associated with C4 and Dolma [1, 2]; their measured performance applies to the specified transformations.

C4-style implementation. We split sentences at whitespace following terminal punctuation or at line breaks, then trim nonempty sentences. Each original three-sentence window receives a lowercase, whitespace-normalized key that preserves punctuation, numbers, and accents. Sources are traversed newest first using the recorded creation, version, and document identifiers. A previously seen window marks only its first sentence for deletion; every original window enters the family-local exact set. Windows are defined before deletion, with no new sliding windows or cross-document windows. The final two sentences have no complete starting window and therefore survive this rule. Repeated windows within a document are eligible. We concatenate retained bodies in traversal order, omit titles, and apply no extra quality or short-document filter. This first-sentence variant differs from removing a repeated three-sentence span in full.

Dolma-style implementation. We split only at original newlines, ignore empty lines, trim paragraphs, and match lowercase keys with collapsed whitespace. Newest-first traversal keeps the first matching paragraph, including within-document deduplication. Text without newlines remains a single paragraph. The exact set resets per family, so this implementation introduces no Bloom-filter false positives. Retained bodies are concatenated with empty titles.

Table 2: CC-News matched budgets. Treated families are counted out of 150; all test families remain in the metric denominators. Removal matches the nominal budget to integer-word rounding. Brackets give 95% CIs; loss and outdated retention are percentages.

Budget	Method	Treated	Lost / inventory; rate [95% CI]	Outdated retained / total; rate [95% CI]
10.00%	Latest	21/150	11/4483; 0.25% [0.07%, 0.48%]	9/10; 90.00% [66.67%, 100.00%]
10.00%	Longest	23/150	8/4483; 0.18% [0.04%, 0.37%]	9/10; 90.00% [66.67%, 100.00%]
10.00%	Sentence-Exact	29/150	4/4483; 0.09% [0.00%, 0.20%]	10/10; 100.00% [100.00%, 100.00%]
10.00%	Sub-doc merge	24/150	3/4483; 0.07% [0.00%, 0.15%]	9/10; 90.00% [66.67%, 100.00%]
10.00%	C4-style	38/150	4/4483; 0.09% [0.00%, 0.20%]	10/10; 100.00% [100.00%, 100.00%]
10.00%	Dolma-style	37/150	5/4483; 0.11% [0.02%, 0.24%]	10/10; 100.00% [100.00%, 100.00%]
20.00%	Latest	42/150	18/4483; 0.40% [0.18%, 0.68%]	9/10; 90.00% [66.67%, 100.00%]
20.00%	Longest	43/150	13/4483; 0.29% [0.11%, 0.49%]	9/10; 90.00% [66.67%, 100.00%]
20.00%	Sentence-Exact	53/150	13/4483; 0.29% [0.09%, 0.54%]	10/10; 100.00% [100.00%, 100.00%]
20.00%	Sub-doc merge	45/150	6/4483; 0.13% [0.02%, 0.27%]	9/10; 90.00% [66.67%, 100.00%]
20.00%	C4-style	76/150	14/4483; 0.31% [0.11%, 0.57%]	10/10; 100.00% [100.00%, 100.00%]
20.00%	Dolma-style	64/150	13/4483; 0.29% [0.09%, 0.54%]	10/10; 100.00% [100.00%, 100.00%]

Table 3: CC-News paired loss differences in percentage points, left minus right. All comparisons use the same 4,483-fact test inventory; brackets give 95% CIs.

Budget	Comparison	Difference [95% CI]
10.00%	Latest − Sub-doc merge	0.18 [0.02, 0.39]
10.00%	Longest − Sub-doc merge	0.11 [-0.02, 0.30]
10.00%	Sentence-Exact − Sub-doc merge	0.02 [-0.07, 0.12]
20.00%	Latest − Sub-doc merge	0.27 [0.09, 0.51]
20.00%	Longest − Sub-doc merge	0.16 [-0.02, 0.36]
20.00%	Sentence-Exact − Sub-doc merge	0.16 [-0.02, 0.38]
10.00%	Latest − Sentence-Exact	0.16 [-0.04, 0.38]
10.00%	Longest − Sentence-Exact	0.09 [-0.07, 0.28]
20.00%	Latest − Sentence-Exact	0.11 [-0.19, 0.41]
20.00%	Longest − Sentence-Exact	0.00 [-0.29, 0.26]

Table 4: CC-News C4-style and Dolma-style paired loss differences in percentage points; brackets give 95% CIs.

Budget	Comparison	Difference [95% CI]
10.00%	C4-style − Latest	-0.16 [-0.38, 0.04]
10.00%	C4-style − Sentence-Exact	0.00 [0.00, 0.00]
10.00%	C4-style − Sub-doc merge	0.02 [-0.07, 0.12]
10.00%	Dolma-style − Latest	-0.13 [-0.37, 0.07]
10.00%	Dolma-style − Sentence-Exact	0.02 [0.00, 0.07]
10.00%	Dolma-style − Sub-doc merge	0.04 [-0.06, 0.16]
20.00%	C4-style − Latest	-0.09 [-0.39, 0.23]
20.00%	C4-style − Sentence-Exact	0.02 [0.00, 0.07]
20.00%	C4-style − Sub-doc merge	0.18 [-0.02, 0.44]
20.00%	Dolma-style − Latest	-0.11 [-0.41, 0.19]
20.00%	Dolma-style − Sentence-Exact	0.00 [0.00, 0.00]
20.00%	Dolma-style − Sub-doc merge	0.16 [-0.02, 0.38]

The experiment transfers the paragraph-matching mechanism, with an exact set replacing the corpus-scale Bloom filter and without the full official cleaning pipeline.

Frozen comparison. The earlier matched-budget comparison reuses the historical split of 59 development and 60 test AP families, and 57 development and 58 test FW families. Test denominators are 2,390 non-outdated and 40 outdated AP facts, versus 1,230 and one in FW. New judgments cover 231 distinct texts and 7,190 fact–text cells using `gpt-6.1-sol`, with no missing cells. The fact inventory stays fixed; judging shuffles document letters and hides policy names. Development selects activation rates using word counts, and the test controller matches integer word targets without consulting fact labels. The removal baselines are 87,555 AP words and 35,141 FW words: AP targets are 8,756, 17,511, and 26,266 words; FW targets are 3,514 and 7,028 words. Table 5 places both adaptations alongside the principal controls on this previously observed split.

Loss and outdated retention. Across the AP budgets, C4-style loses 0/2,390 facts and Dolma-style 1/2,390, versus Sentence-Exact's 2/2,390. On FW, C4-style loses 14/1,230 facts (1.14%) at 10% removal and 41/1,230 (3.33%) at 20%. Its differences from Sentence-Exact are +0.16 pp [95% CI: −0.17, 0.58] and +0.73 pp [−2.20, 3.70]. Define the recovered share of Sentence-Exact's benefit as $(\ell_{\text{Latest}} - \ell_m)/(\ell_{\text{Latest}} - \ell_{\text{Sentence}})$, using lost-fact counts ℓ. For C4-style these FW shares are $(23-14)/(23-12) = 81.82\%$ and $(63-41)/(63-32) = 70.97\%$. The adaptations retain 38–39/40 outdated AP facts and 1/1 in FW;

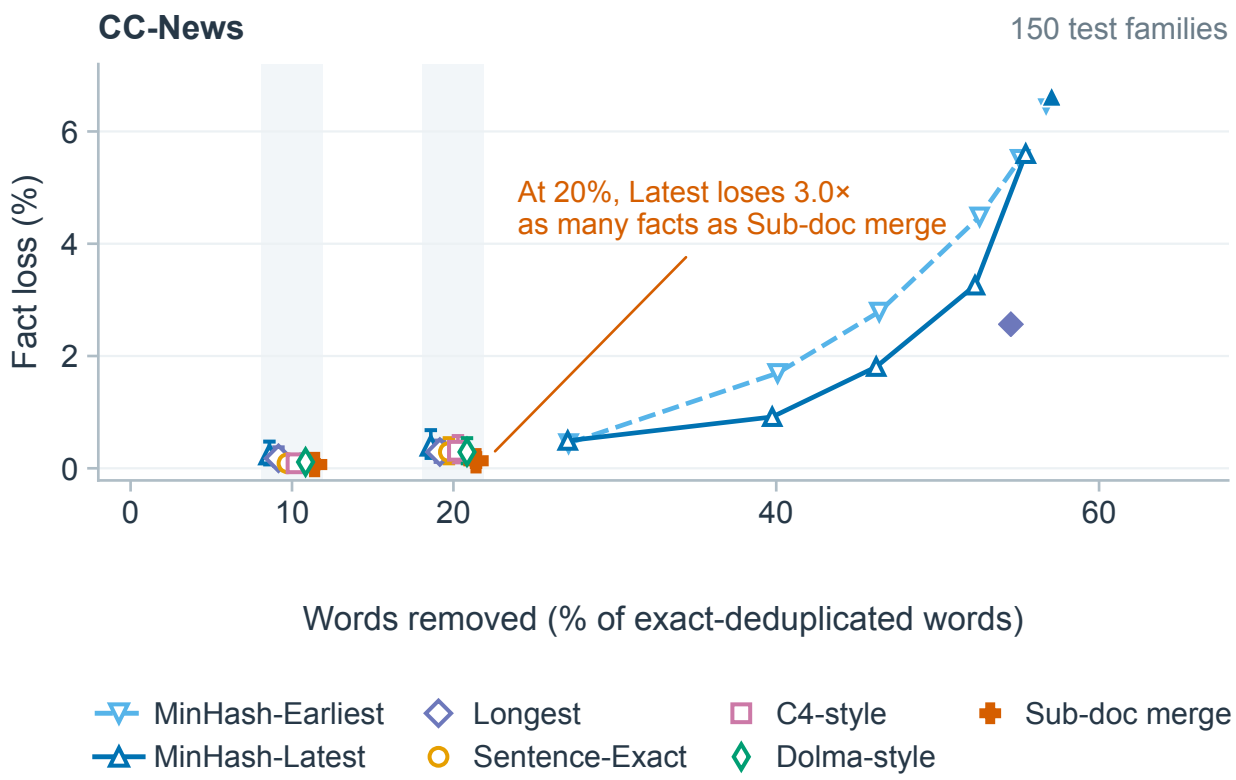


Figure 1: CC-News removal–loss trade-offs on the 150 test families (140,636 exact-deduplicated words; 4,483 non-outdated facts). Curves connect observed document-threshold points; filled endpoints keep one document per family. Matched-budget symbols show the six methods at 10% and 20%, with 95% CIs. Their horizontal offsets serve only to separate symbols; the vertical guides mark the actual nominal budgets. Curves and matched-budget points represent distinct operating procedures.

that single FW fact provides little basis for comparing outdated retention.

Repeated-judgment sensitivity. Support status differs in 22/1,321 cells (1.67%) across the 55 texts shared with the earlier judging pool. Reusing the earlier labels for those identical bodies leaves C4-style unchanged, but moves Dolma-style's FW losses from 19 to 20/1,230 at 10% removal and from 35 to 57/1,230 at 20%. Its recovered benefit shares change from 36.36% to 27.27% and from 90.32% to 19.35%, respectively. After reuse, Dolma-style minus Sentence-Exact is +0.65 pp [95% CI: −1.00, 2.55] and +2.03 pp [−1.57, 6.95]. The primary table retains the original judgments; no human adjudication determined which repeated labels were correct.

Allocation sensitivity. Five fixed family orders, seeded 20261011 through 20261015, vary only the word-budget assignment while preserving parameters and candidate texts. Table 6 reports the resulting minimum and maximum differences from Sentence-Exact. Both FW adaptations change direction across orders, making the recovered-benefit description specific to the primary allocation; these ranges are sensitivity summaries, not CIs.

S5 Retrieval QA: Earlier Evaluations and Robustness

Deduplication changes which previously correct questions remain correct in the fixed retrieval-QA pipeline. We report each transition rate as the fraction of Keep-all-correct questions that become incorrect, alongside total accuracy; improvements on other questions can mask these losses in average accuracy.

Historical evaluation. The historical run samples 600 non-outdated facts per source and generates 507 AP and 558 FW questions. Excluding all questions in blocked service batches across every policy removes nine AP and six FW questions, leaving 498 and 552. The evaluable total is 1,050 questions and 8,400 policy–question units across eight policies, all with successful answers and scores. Questions cover 396 AP and 407 FW families. We generate questions and answers with `gpt-6-astra` at medium reasoning and score answer equivalence with the same model at low reasoning. BM25 indexes lowercase Unicode terms from bodies, with $k_1 = 1.2$, $b = 0.75$, nonoverlapping 120-word chunks, and the top five passages; titles are shown with passages but excluded from indexing. Identical question–context inputs reuse answers and scores. Safety-batch

Table 5: Principal controls and industry-style adaptations at matched budgets. Latest and Longest use document-component representatives. Fact columns give counts, percentage rates, and 95% CIs. Treated-family counts distinguish allocation coverage from the full test denominator.

Panel / budget	Method	Treated	Lost / inventory; rate [95% CI]	Outdated retained / total; rate [95% CI]
AP/10.00%	Latest	8/60	8/2390; 0.33% [0.04%, 0.72%]	37/40; 92.50% [62.50%, 100.00%]
AP/10.00%	Longest	8/60	8/2390; 0.33% [0.04%, 0.72%]	37/40; 92.50% [62.50%, 100.00%]
AP/10.00%	Sentence-Exact	11/60	2/2390; 0.08% [0.00%, 0.26%]	40/40; 100.00% [100.00%, 100.00%]
AP/10.00%	Sub-doc merge	12/60	2/2390; 0.08% [0.00%, 0.21%]	37/40; 92.50% [62.50%, 100.00%]
AP/10.00%	C4-style	15/60	0/2390; 0.00% [0.00%, 0.00%]	39/40; 97.50% [87.50%, 100.00%]
AP/10.00%	Dolma-style	11/60	1/2390; 0.04% [0.00%, 0.13%]	39/40; 97.50% [87.50%, 100.00%]
AP/20.00%	Latest	21/60	10/2390; 0.42% [0.12%, 0.81%]	37/40; 92.50% [62.50%, 100.00%]
AP/20.00%	Longest	20/60	12/2390; 0.50% [0.16%, 0.93%]	36/40; 90.00% [56.25%, 100.00%]
AP/20.00%	Sentence-Exact	26/60	2/2390; 0.08% [0.00%, 0.26%]	40/40; 100.00% [100.00%, 100.00%]
AP/20.00%	Sub-doc merge	22/60	4/2390; 0.17% [0.00%, 0.38%]	36/40; 90.00% [56.25%, 100.00%]
AP/20.00%	C4-style	31/60	0/2390; 0.00% [0.00%, 0.00%]	38/40; 95.00% [79.99%, 100.00%]
AP/20.00%	Dolma-style	26/60	1/2390; 0.04% [0.00%, 0.13%]	39/40; 97.50% [87.50%, 100.00%]
AP/30.00%	Latest	26/60	10/2390; 0.42% [0.16%, 0.73%]	34/40; 85.00% [44.00%, 98.70%]
AP/30.00%	Longest	25/60	13/2390; 0.54% [0.20%, 0.97%]	34/40; 85.00% [44.00%, 98.70%]
AP/30.00%	Sentence-Exact	30/60	2/2390; 0.08% [0.00%, 0.26%]	40/40; 100.00% [100.00%, 100.00%]
AP/30.00%	Sub-doc merge	28/60	4/2390; 0.17% [0.00%, 0.38%]	35/40; 87.50% [50.00%, 100.00%]
AP/30.00%	C4-style	36/60	0/2390; 0.00% [0.00%, 0.00%]	38/40; 95.00% [79.99%, 100.00%]
AP/30.00%	Dolma-style	31/60	1/2390; 0.04% [0.00%, 0.13%]	39/40; 97.50% [87.50%, 100.00%]
FW/10.00%	Latest	7/58	23/1230; 1.87% [0.38%, 3.97%]	0/1; 0.00% [0.00%, 0.00%]
FW/10.00%	Longest	11/58	27/1230; 2.20% [0.63%, 4.11%]	1/1; 100.00% [100.00%, 100.00%]
FW/10.00%	Sentence-Exact	14/58	12/1230; 0.98% [0.21%, 2.26%]	1/1; 100.00% [100.00%, 100.00%]
FW/10.00%	Sub-doc merge	14/58	15/1230; 1.22% [0.17%, 2.67%]	1/1; 100.00% [100.00%, 100.00%]
FW/10.00%	C4-style	19/58	14/1230; 1.14% [0.29%, 2.50%]	1/1; 100.00% [100.00%, 100.00%]
FW/10.00%	Dolma-style	19/58	19/1230; 1.54% [0.34%, 3.27%]	1/1; 100.00% [100.00%, 100.00%]
FW/20.00%	Latest	18/58	63/1230; 5.12% [2.61%, 8.27%]	0/1; 0.00% [0.00%, 0.00%]
FW/20.00%	Longest	17/58	73/1230; 5.93% [2.20%, 11.28%]	1/1; 100.00% [100.00%, 100.00%]
FW/20.00%	Sentence-Exact	28/58	32/1230; 2.60% [0.93%, 4.92%]	1/1; 100.00% [100.00%, 100.00%]
FW/20.00%	Sub-doc merge	27/58	35/1230; 2.85% [0.99%, 5.44%]	1/1; 100.00% [100.00%, 100.00%]
FW/20.00%	C4-style	37/58	41/1230; 3.33% [1.33%, 5.99%]	1/1; 100.00% [100.00%, 100.00%]
FW/20.00%	Dolma-style	30/58	35/1230; 2.85% [1.16%, 5.11%]	1/1; 100.00% [100.00%, 100.00%]

Table 6: Loss difference from Sentence-Exact across five fixed allocation orders, in percentage points. Negative values favor the listed adaptation. Entries are minimum / maximum, not confidence intervals.

Panel / budget	Method	Difference range
AP/10.00%	C4-style	-0.08 / 0.00
AP/10.00%	Dolma-style	-0.04 / 0.00
AP/20.00%	C4-style	-0.08 / 0.00
AP/20.00%	Dolma-style	-0.04 / 0.00
AP/30.00%	C4-style	-0.08 / -0.08
AP/30.00%	Dolma-style	-0.04 / -0.04
FW/10.00%	C4-style	-0.73 / 1.22
FW/10.00%	Dolma-style	-0.73 / 0.57
FW/20.00%	C4-style	-1.87 / 1.14
FW/20.00%	Dolma-style	-2.28 / 1.22

exclusions and successful question generation define this evaluable population.

Table 7 preserves the historical answers with the harmonized removal coordinates and recomputed intervals in the final numerical record. Keep-all answers 432/498 AP and 476/552 FW questions correctly. Keep-One-Earliest changes 127/432 (29.40%) and 216/476 (45.38%) of these to incorrect answers; Sub-doc merge@0.3 changes 10/432 (2.31%) and 38/476 (7.98%). The policies operate at different removal budgets, so these contrasts describe their observed operating points.

Context-aligned subset. The context check finds 49 retrieved passages from families excluded from the final corpus snapshot. If any policy retrieves such a passage, we exclude that question from every policy, using source identifiers alone. This removes three AP and ten FW questions, leaving 495 and 542. Keep-all denominators become 430 and 467 correct answers (Table 8). Removal uses the final exact-word baselines of 1,672,964 AP and 665,272 FW words; intervals use 2,000 replicates with seed 20261011. This reanalysis reuses historical answers and retains historical BM25 document frequencies and rankings, rather than rerunning retrieval on the final snapshot.

Model and retrieval changes. The robustness sample fixes 100 questions per source: 30 Keep-all-incorrect questions, 35 Keep-all-correct questions with a policy loss, and 35 correct under every policy. Common service-batch exclusions leave 97 AP and 98 FW questions. We compare the reused BM25/Astra answers with

Table 7: Historical answers, with harmonized removal coordinates and recomputed 95% CIs. Correct counts use all evaluable questions; transition counts use Keep-all-correct questions. Removal denominators are 1,672,964 AP and 665,272 FW exact-deduplicated words, not the historical raw-word baselines.

Panel	Policy	Correct / questions	Correct → incorrect; rate	95% CI	Removal
AP	Keep-all	432/498	0/432 = 0.00%	[0.00%, 0.00%]	0.00%
AP	MinHash-Earliest@0.5	435/498	22/432 = 5.09%	[3.10%, 7.26%]	33.60%
AP	MinHash-Latest@0.5	446/498	14/432 = 3.24%	[1.69%, 4.98%]	31.02%
AP	MinHash-Earliest@0.3	423/498	35/432 = 8.10%	[5.42%, 10.82%]	39.74%
AP	MinHash-Latest@0.3	445/498	16/432 = 3.70%	[2.07%, 5.57%]	35.90%
AP	Keep-One-Earliest	333/498	127/432 = 29.40%	[25.11%, 33.78%]	64.20%
AP	Keep-One-Longest	399/498	63/432 = 14.58%	[11.22%, 17.97%]	54.14%
AP	Sub-doc merge@0.3	452/498	10/432 = 2.31%	[0.98%, 3.82%]	30.50%
FW	Keep-all	476/552	0/476 = 0.00%	[0.00%, 0.00%]	0.00%
FW	MinHash-Earliest@0.5	445/552	36/476 = 7.56%	[5.00%, 10.08%]	14.59%
FW	MinHash-Latest@0.5	439/552	40/476 = 8.40%	[5.81%, 11.38%]	14.26%
FW	MinHash-Earliest@0.3	298/552	186/476 = 39.08%	[34.10%, 43.86%]	50.89%
FW	MinHash-Latest@0.3	312/552	171/476 = 35.92%	[31.61%, 40.51%]	49.76%
FW	Keep-One-Earliest	267/552	216/476 = 45.38%	[40.13%, 50.21%]	58.19%
FW	Keep-One-Longest	316/552	169/476 = 35.50%	[30.91%, 39.71%]	53.72%
FW	Sub-doc merge@0.3	448/552	38/476 = 7.98%	[5.70%, 10.41%]	30.84%

Table 8: Context-aligned subset of the historical QA answers. Counts and intervals are paired across policies. Removal coordinates are the harmonized values in Table 7.

Panel	Policy	Correct / questions	Correct → incorrect; rate	95% CI
AP	Keep-all	430/495	0/430 = 0.00%	[0.00%, 0.00%]
AP	MinHash-Earliest@0.5	433/495	22/430 = 5.12%	[3.10%, 7.25%]
AP	MinHash-Latest@0.5	444/495	14/430 = 3.26%	[1.67%, 4.95%]
AP	MinHash-Earliest@0.3	421/495	35/430 = 8.14%	[5.50%, 11.22%]
AP	MinHash-Latest@0.3	443/495	16/430 = 3.72%	[2.06%, 5.57%]
AP	Keep-One-Earliest	331/495	127/430 = 29.53%	[25.22%, 34.19%]
AP	Keep-One-Longest	397/495	63/430 = 14.65%	[11.21%, 18.24%]
AP	Sub-doc merge@0.3	450/495	10/430 = 2.33%	[0.96%, 3.80%]
FW	Keep-all	467/542	0/467 = 0.00%	[0.00%, 0.00%]
FW	MinHash-Earliest@0.5	436/542	36/467 = 7.71%	[5.18%, 10.40%]
FW	MinHash-Latest@0.5	430/542	40/467 = 8.57%	[5.86%, 11.28%]
FW	MinHash-Earliest@0.3	289/542	186/467 = 39.83%	[34.97%, 44.75%]
FW	MinHash-Latest@0.3	305/542	169/467 = 36.19%	[31.55%, 40.61%]
FW	Keep-One-Earliest	260/542	214/467 = 45.82%	[41.05%, 50.64%]
FW	Keep-One-Longest	309/542	167/467 = 35.76%	[31.18%, 40.37%]
FW	Sub-doc merge@0.3	439/542	38/467 = 8.14%	[5.82%, 10.47%]

`gpt-6.1-sol` answering and scoring under the same retrieval, then change retrieval to TF-IDF with nonoverlapping 240-word chunks and top-five passages. TF-IDF uses unigrams and bigrams, sublinear term frequency, smoothed inverse document frequency, and cosine similarity. In this earlier robustness check, a dense retriever could not be run, so TF-IDF served as the alternative; the main evaluation in Section 5.5 later added a dense retriever. Changing the model also changes the scorer, while changing retrieval also changes chunk length.

Table 9 gives raw transition counts and design-weighted rates. Weights are the original stratum ratios N_h/n_h; the rate is $\sum_i w_i \mathbf{1}\{\text{Keep-all correct, policy incorrect}\}_i / \sum_i w_i \mathbf{1}\{\text{Keep-all correct}\}_i$. Hence the displayed percentage generally differs from the adjacent integer ratio. Approximate 95% CIs use 2,000 paired family-bootstrap replicates with stratum-weight calibration. Keep-One-Earliest remains at the high-loss end across configurations, while Sub-doc merge@0.3 remains lower. The direction persists despite shifts in estimates and baseline-correct denominators.

Oracle and scoring diagnostics. Giving the answerer the shortest source labeled as supporting each gold fact produces 94/97 correct AP answers and 90/98 correct FW answers. The answerer sees the source, not the gold answer. This oracle helps separate failures with deleted known support, retained support missed by retrieval, supported contexts answered incorrectly, and unresolved oracle failures. Known-support deletion refers to the question's original family and therefore leaves possible support elsewhere in the retained corpus unresolved.

A nonblind secondary LLM review covers 60 failed questions and 20 successful controls, with each of the 80 distinct questions reviewed once. Agreement with the main labels is 79/80 for answer correctness and 79/80 for $\sum_i w_i \mathbf{1}\{\text{Keep-all correct, policy incorrect}\}_i / \sum_i w_i \mathbf{1}\{\text{Keep-all correct}\}_i$. reviewers flag question ambiguity or gold-answer scope in 16/80. The review preserves all primary scores. These are automated model checks, not human annotation or inter-annotator agreement.

Table 9: Robustness on the common evaluable sample. Each entry gives raw correct-to-incorrect counts / that configuration's Keep-all-correct count, followed by the weighted percentage and its 95% CI. Astra and Sol abbreviate the model configurations described in the text.

Panel	Policy	BM25 / Astra	BM25 / Sol	TF-IDF / Sol
AP	MinHash-Earliest@0.5	5/67; 5.38% [1.41%, 10.16%]	4/67; 4.33% [0.99%, 8.64%]	5/77; 4.68% [1.08%, 8.98%]
AP	MinHash-Latest@0.5	5/67; 5.38% [1.22%, 10.16%]	4/67; 4.33% [0.96%, 8.92%]	3/77; 2.60% [0.00%, 6.22%]
AP	MinHash-Earliest@0.3	10/67; 10.76% [5.37%, 16.52%]	9/67; 9.73% [4.39%, 15.61%]	10/77; 9.88% [4.79%, 15.65%]
AP	MinHash-Latest@0.3	6/67; 6.45% [2.03%, 11.49%]	5/67; 5.41% [1.21%, 10.35%]	5/77; 4.15% [1.01%, 8.33%]
AP	Keep-One-Earliest	30/67; 32.27% [27.70%, 35.58%]	29/67; 31.36% [26.67%, 35.13%]	33/77; 32.23% [27.38%, 36.52%]
AP	Keep-One-Longest	12/67; 12.91% [7.13%, 19.25%]	11/67; 11.90% [6.34%, 18.02%]	12/77; 10.91% [5.29%, 16.90%]
AP	Sub-doc merge@0.3	4/67; 4.30% [0.96%, 8.60%]	3/67; 3.24% [0.00%, 7.32%]	2/77; 1.56% [0.00%, 4.24%]
FW	MinHash-Earliest@0.5	3/69; 6.61% [0.00%, 14.49%]	3/68; 6.76% [0.00%, 14.85%]	6/76; 7.98% [1.43%, 15.94%]
FW	MinHash-Latest@0.5	6/69; 13.21% [4.83%, 23.53%]	5/68; 11.26% [3.46%, 21.25%]	7/76; 11.71% [4.05%, 21.04%]
FW	MinHash-Earliest@0.3	16/69; 35.24% [22.02%, 48.31%]	15/68; 33.78% [20.55%, 46.88%]	19/76; 35.70% [22.88%, 48.85%]
FW	MinHash-Latest@0.3	17/69; 37.44% [24.29%, 49.28%]	17/68; 38.28% [25.34%, 50.42%]	20/76; 37.83% [25.15%, 49.95%]
FW	Keep-One-Earliest	18/69; 39.64% [26.96%, 52.19%]	17/68; 38.28% [24.96%, 50.65%]	24/76; 41.54% [29.17%, 54.00%]
FW	Keep-One-Longest	12/69; 26.43% [14.39%, 37.44%]	12/68; 27.02% [14.98%, 39.16%]	16/76; 27.69% [16.21%, 38.90%]
FW	Sub-doc merge@0.3	4/69; 8.81% [2.14%, 17.48%]	4/68; 9.01% [2.19%, 17.88%]	5/76; 7.45% [1.04%, 15.39%]

S6 Training Experiments and Failed Positive Controls

Both training checks failed their learning positive controls. They therefore leave the connection between corpus fact loss and learned knowledge unresolved.

Continued pretraining. We continue full-parameter pretraining of Qwen2.5-0.5B on the AP and FW corpora, comparing the exact-deduplicated Keep-all corpus, Keep-One-Earliest, MinHash-Latest@0.5, and Sentence-Exact activated by word counts to approximate the latest-representative removal rate. Every retained token receives the same number of epochs. Training uses AdamW with learning rate 5×10^{-5}, sequence length 512, microbatch size two, and gradient accumulation eight. Closed-book evaluation uses questions with fixed generic format examples, no retrieved passages, greedy decoding, and at most 48 generated tokens; `gpt-6-astra` scores answer equivalence, with a rule-matching check alongside it. Equal epochs imply different training-token and update counts across policies.

The positive-control check fixes 200 questions, 100 per source. At epochs 1, 3, and 6, Keep-all must improve semantic accuracy over the base model by at least five percentage points and have a paired 95% CI lower bound above zero. Against the pilot base score of 8/200, the checkpoints score 13/200, 12/200, and 10/200; all fail (Table 10). The fallback to six epochs for exploratory policy comparisons was recorded after the third-epoch failure and before observing the sixth-epoch result.

The final evaluation covers 1,037 questions with two seeds per trained policy (Table 11). Keep-all scores 131/2,074 predictions correctly (6.32%), compared with the base model's 68/1,037 (6.56%); its change is −0.24 pp [95% CI: −1.73, 1.33]. Policy-minus-Keep-all differences are +0.10 pp [−1.11, 1.27] for Keep-One-Earliest, +0.72 pp [−0.44, 1.89] for MinHash-Latest, and −0.77 pp [−1.82, 0.34] for Sentence-Exact. Final scoring uses a unified BF16 inference path, whereas the pilot used FP32 weights with BF16 autocast; the recorded pilot gate is preserved. These policy comparisons remain exploratory because Keep-all failed to demonstrate the required learning gain.

Knowledge-injection follow-up. The follow-up protocol specifies a stronger Qwen2.5 base model with QLoRA and training on each retained document plus four paraphrases. It targets the families containing the aligned QA questions and requires Keep-all to pass a 200-question learning check before any policy comparison. The run used Qwen2.5-7B with 4-bit QLoRA. Each of the 2,660 retained source texts received four paraphrases (10,640 in total); re-judging 20 audited documents against their paraphrases changed none of 2,808 support labels.

The reported base score is 40/200. Three training lengths (1–3 epochs) produce 50/200, 56/200, and 52/200 correct answers. All fall below the required 70/200, corresponding to a 15-percentage-point improvement over the base. The best observed gain is eight percentage points. These results record a failed positive control; the other-policy comparison is gated on a learning improvement that these runs did not achieve.

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Table 10: Learning checks for continued pretraining. Changes are percentage points with 95% CIs. None passes the required gain and positive lower-bound gate.

Epoch	Base correct	Keep-all correct	Change [95% CI]	Gate
1	8/200	13/200	+2.50 [-0.51, 6.00]	Fail
3	8/200	12/200	+2.00 [-1.49, 5.50]	Fail
6	8/200	10/200	+1.00 [-2.46, 4.48]	Fail

Table 11: Final semantic accuracy after continued pretraining. Trained-policy counts pool two seeds over the same 1,037 questions. Brackets give 95% CIs from resampling question families while carrying both seeds together.

Model / policy	Seeds	Correct / predictions	Accuracy [95% CI]
Base	1	68/1037	6.56% [5.10, 8.16]
Keep-all	2	131/2074	6.32% [5.18, 7.63]
Keep-One-Earliest	2	133/2074	6.41% [5.14, 7.77]
MinHash-Latest@0.5	2	146/2074	7.04% [5.75, 8.40]
Sentence-Exact	2	115/2074	5.54% [4.37, 6.88]